Exercises: Advanced Querying

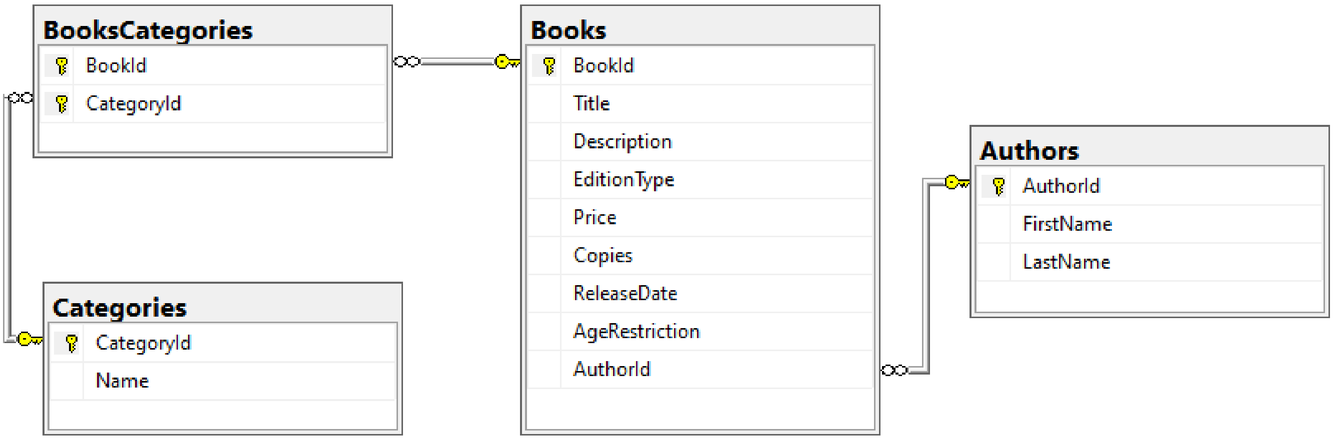
You can check your solutions in Judge

BookShop System

For the following tasks, use the **BookShop** database. You can download the complete project or create it, but you should still use the pre-defined **`Seed()`** method from the project to have the same **sample** data.

1. Book Shop Database

You must create a **database** for a **book shop system**. It should look like this:



Constraints

Your **namespaces** should be:

- **`BookShop`** – for your **`StartUp`** class
- **`BookShop.Data`** – for your **`DbContext`**
- **`BookShop.Models`** – for your models
- **`BookShop.Models.Enums`** – for your models

Your **models** should be:

- **`BookShopContext`** – your **`DbContext`**
- **`Author`**
  - **`AuthorId`**
  - **`FirstName`** (up to **`50`** characters, unicode, not required)
  - **`LastName`** (up to **`50`** characters, unicode)
- **`Book`**
  - **`BookId`**
  - **`Title`** (up to **`50`** characters, unicode)
  - **`Description`** (up to **`1000`** characters, unicode)
  - **`ReleaseDate`** (not required)
  - **`Copies`** (an integer)
  - **`Price`**
  - **`EditionType`** – enum (**`Normal`**, **`Promo`**, **`Gold`**)
  - **`AgeRestriction`** – enum (**`Minor`**, **`Teen`**, **`Adult`**)

    - `Author`
    - `BookCategories`
- `Category`
    - `CategoryId`
    - `Name` (up to **50** characters, unicode)
    - `CategoryBooks`
- `BookCategory` – mapping entity

For the following tasks, you will be creating methods that accept a **`BookShopContext`** as a parameter and use it to run some queries. Create those methods inside your **`StartUp`** class and upload your whole solution to **Judge**.

2. Age Restriction

NOTE: You will need method **`public static string GetBooksByAgeRestriction(BookShopContext context, string command)`** and **`public StartUp`** class.

Return in a **single string** all book **titles**, each on a **new line,** that have **an age restriction**, equal to the **given command**. Order the titles **alphabetically**.

Read **input** from the console in your **main method** and call your **method** with the **necessary arguments**. Print the **returned string** to the console. **Ignore** the casing of the input.

Example

Input	Output
miNor	A Confederacy of Dunces A Farewell to Arms A Handful of Dust …
teEN	A Passage to India A Scanner Darkly A Swiftly Tilting Planet …

3. Golden Books

NOTE: You will need **method `public static string GetGoldenBooks(BookShopContext context)`** and **`public StartUp`** class.

Return in a **single** string the **titles of the golden edition books** that have **less than `5000` copies**, each on a **new line**. Order them by **`BookId`** ascending.

Call the **`GetGoldenBooks(BookShopContext context)`** method in your **`Main()`** and print the returned string to the console.

Example

Output
Behold the Man

`Bury My Heart at Wounded Knee`
`The Cricket on the Hearth`
…

4. Books by Price

<u>**NOTE**</u>: You will need method **`public static string GetBooksByPrice(BookShopContext context)`** and **`public StartUp`** class.

Return in a single string all **titles and prices of books** with a **price higher than `40`**, each on a **new row** in the **format** given below. Order them by **price** descending.

Example

Output

`O Pioneers! - $49.90`
`That Hideous Strength - $48.63`
`A Handful of Dust - $48.63`
…

5. Not Released In

<u>**NOTE**</u>: You will need method **`public static string GetBooksNotReleasedIn(BookShopContext context, int year)`** and **`public StartUp`** class.

Return in a **single** string with all **titles of books** that are **NOT released** in a given year. Order them by **`bookId`** ascending.

Example

Input	Output

`2000`	`Absalom`
	`After Many a Summer Dies the Swan`
	`Ah`
	…

`1998`	`Ah`
	`Wilderness!`
	`Alien CornÂ (play)`
	…

6. Book Titles by Category

<u>**NOTE**</u>: You will need method **`public static string GetBooksByCategory(BookShopContext context, string input)`** and **`public StartUp`** class.

Return in a single string the **titles of books** by a given **list of categories**. The list of **categories** will be given in a single line separated by one or more spaces. Ignore casing. Order by **title** alphabetically.

Example

Input	Output
horror mystery drama	A Fanatic Heart A Farewell to Arms A Glass of Blessings …

7. Released Before Date

NOTE: You will need method **public static string GetBooksReleasedBefore(BookShopContext context, string date)** and **public StartUp** class.

Return **the title, edition type** and **price** of **all books** that are **released before a given date.** The date will be a string in the format **"dd-MM-yyyy".**

Return all of the rows in a single string, ordered by **release date (descending)**.

Example

Input	Output
12-04-1992	If I Forget Thee Jerusalem - Gold - $33.21 Oh! To be in England - Normal - $46.67 The Monkey's Raincoat - Normal - $46.93 …
30-12-1989	A Fanatic Heart - Normal - $9.41 The Curious Incident of the Dog in the Night-Time - Normal - $23.41 The Other Side of Silence - Gold - $46.26 …

8. Author Search

NOTE: You will need method **public static string GetAuthorNamesEndingIn(BookShopContext context, string input)** and **public StartUp** class.

Return the **full names** of **authors**, whose **first name** ends with a **given string**.

Return all **names** in a **single string**, each on a **new row**, ordered alphabetically.

Example

Input	Output
e	George Powell Jane Ortiz
dy	Randy Morales

9. Book Search

<u>NOTE</u>: You will need method **`public static string GetBookTitlesContaining(BookShopContext context, string input)` and `public StartUp`** class.

Return the **titles** of **the book**, which contain a **given string**. Ignore casing.

Return all **titles** in a **single string**, each on a **new row**, ordered alphabetically.

Example

Input	Output
sK	A Catskill Eagle The Daffodil Sky The Skull Beneath the Skin
WOR	Great Work of Time Terrible Swift Sword

10. Book Search by Author

<u>NOTE</u>: You will need method **`public static string GetBooksByAuthor(BookShopContext context, string input)`** and **`public StartUp`** class.

Return **all titles of books and their authors' names** for books, which are written by authors whose last names **start with the given string**.

Return a single string with each title on a new row. **Ignore** casing. Order by **`BookId`** ascending.

Example

Input	Output
R	A Handful of Dust (Bozhidara Rysinova) Have His Carcase (Bozhidara Rysinova) The Heart Is a Lonely Hunter (Bozhidara Rysinova) …
po	Postern of Fate (Stanko Popov) Precious Bane (Stanko Popov) The Proper Study (Stanko Popov) …

11. Count Books

<u>NOTE</u>: You will need method **`public static int CountBooks(BookShopContext context, int lengthCheck)`** and **`public StartUp`** class.

Return **the number of books,** which have a **title longer than the number** given as an input.

Example

Input	Output	Comments
12	169	There are 169 books with longer title than 12 symbols
40	2	There are 2 books with longer title than 40 symbols

12. Total Book Copies

NOTE: You will need method **`public static string CountCopiesByAuthor(BookShopContext context)`** and **`public StartUp`** class.

Return the **total number of book copies for each author**. Order the results **descending by total book copies**.

Return all results in a **single string**, each on a **new line**.

Example

Output
Stanko Popov - 117778 Lyubov Ivanova - 107391 Jane Ortiz – 103673 …

13. Profit by Category

NOTE: You will need method **`public static string GetTotalProfitByCategory(BookShopContext context)`** and **`public StartUp`** class.

Return the **total profit of all books by category**. Profit for a book can be calculated by multiplying its **number of copies** by the **price per single book**. Order the results by **descending by total profit** for a category and **ascending by category name**. Print the total profit formatted to the **second digit**.

Example

Output
Art $6428917.79 Fantasy $5291439.71 Adventure $5153920.77 Children's $4809746.22 …

14. Most Recent Books

NOTE: You will need method **`public static string GetMostRecentBooks(BookShopContext context)`** and **`public StartUp`** class.

Get the most recent books by categories. The **categories** should be ordered by **name alphabetically**. Only take the **top 3** most recent books from each category – ordered by **release date** (descending). **Select** and **print** the **category name** and for each **book** – its **title** and **release year**.

Example

Output
--Action Brandy ofthe Damned (2015) Bonjour Tristesse (2013) By Grand Central Station I Sat Down and Wept (2010) --Adventure The Cricket on the Hearth (2013) Dance Dance Dance (2002) Cover Her Face (2000) …

15. Increase Prices

__NOTE__: You will need method **`public static void IncreasePrices(BookShopContext context)`** and **`public StartUp`** class.

Increase the prices of all books released before `2010` by `5`.

16. Remove Books

__NOTE__: You will need method **`public static int RemoveBooks(BookShopContext context)`** and **`public StartUp`** class.

Remove all **books**, which have less than **`4200` copies**. Return an **`int`** - the **number of books that were deleted** from the database.

Example

Output
34